

DESIGN AND FABRICATION OF AN ELECTRIC GENERATOR USING WASTE MATERIALS

Introduction

Driven by the need for sustainable energy and effective waste management, our team designed and built an electric generator using waste materials. This project addresses the challenge of converting solid waste into a valuable energy source, reducing landfill waste, and minimizing harmful emissions.



Objectives

- Develop an electric generator using waste materials.
- Explore the potential of waste as a sustainable energy source.
- Minimize landfill waste and convert it into usable energy.
- Implement emission treatment techniques to reduce greenhouse gases.

Methodology

1. **Waste Collection and Preparation:** We collected and sorted waste materials (plastics, rubber, wood, and paper).
2. **Combustion Process:** Waste materials were burned in a controlled environment to produce radiation.



Before



After

3. **Energy Conversion:** Solar panels converted the radiation into electrical energy.
4. **Emission Treatment:** Implemented techniques to minimize the release of greenhouse gases.
5. **Energy Storage:** The generated electricity was stored in batteries for later use.



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Key Components

- Solar Panels
- Waste Incineration Chamber
- Gas Treatment System
- Energy Storage System (Batteries)

Results and Discussion

- Successfully converted waste materials into electricity.
- Demonstrated the potential of reducing landfill waste through energy conversion.
- Identified combustion parameters for optimal energy generation.



Water Treatment for re-purification

Conclusion

This project demonstrates the viability of generating electricity from waste materials, offering a sustainable solution for energy production and waste reduction. Future work will focus on system optimization for increased

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